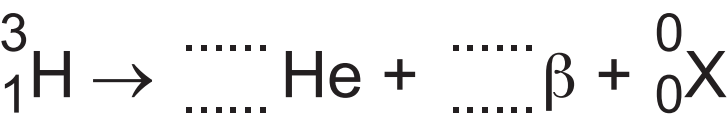


6 The nuclide ${}^3_1\text{H}$ is an isotope of hydrogen that is called tritium.

- (a) (i) Determine the numbers of protons, neutrons and electrons in a neutral atom of tritium.
- number of protons =
number of neutrons =
number of electrons = [2]
- (ii) Draw a labelled diagram to represent a simple model of the arrangement of the protons, neutrons and electrons in a tritium atom.

[2]

- (b) Tritium is radioactive and undergoes β^- decay to form an isotope of helium (He). Gamma radiation is not emitted during this decay.
- (i) Complete the equation to represent the radioactive decay of tritium.



[2]

- (ii) State the name of particle X.

..... [1]

- (c) Determine the quark composition of a tritium nucleus.

.....
..... [2]